

CubeSat

A CubeSat is a class of miniaturized satellite and often use commercial off-the-shelf (COTS) components for their electronics and structure. [<https://en.wikipedia.org/wiki/CubeSat>]

Form Factor:

10cm³

<2kg

CubeSats are deployed into orbit from the International Space Station, or launched as secondary payloads on a launch vehicle.

PURPOSES

to promote and develop the skills necessary for the design, manufacture, and testing of small satellites intended for low Earth orbit (LEO) that perform scientific research and explore new space technologies.

[<https://www.sciencelearn.org.nz/resources/3156-what-is-a-cubesat>,
<https://www.nasa.gov/what-are-smallsats-and-cubesats/>]

- 1) Conducting scientific experiments in space.
- 2) Testing new instruments, components and collecting data.
- 3) provide hands-on experience for students and young engineers.
- 4) enable amateur radio enthusiasts to communicate globally.
- 5) prototyping / technology validation for larger space missions.

CHALLENGES

1) **Orbital Debris**

<https://www.wilsoncenter.org/blog-post/rise-cubesats-opportunities-and-challenges>

To mitigate this, CubeSats should have tracking features and deorbit plans after their mission to prevent long-term debris accumulation.

2) **Short Lifetimes**

3) **Limited Resources & Payload:** due to size constraints

4) **Propulsion and Trajectory Changes:** lack onboard propulsion systems

5) **Communication:** Maintaining reliable communication over vast distances is a challenge.

6) **Radiation Exposure:** Radiation-hardened components are essential for reliability.

CONCLUSION

Their small size, flexibility, and affordability make them valuable tools for scientific research, education, and technology advancement. [<https://www.asc-csa.gc.ca/eng/satellites/cubesat/what-is-a-cubesat.asp>]